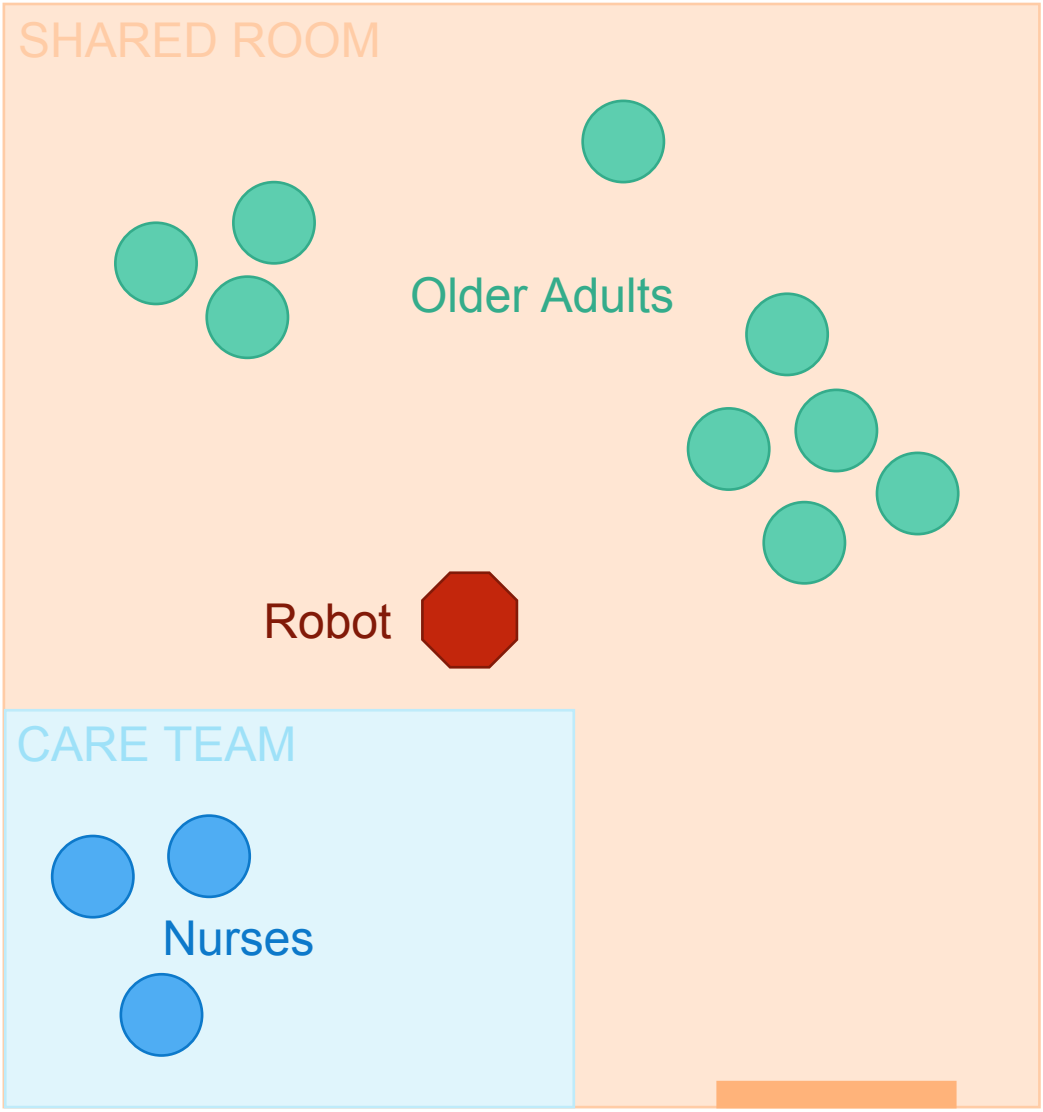
Scenario for human-robot interactions

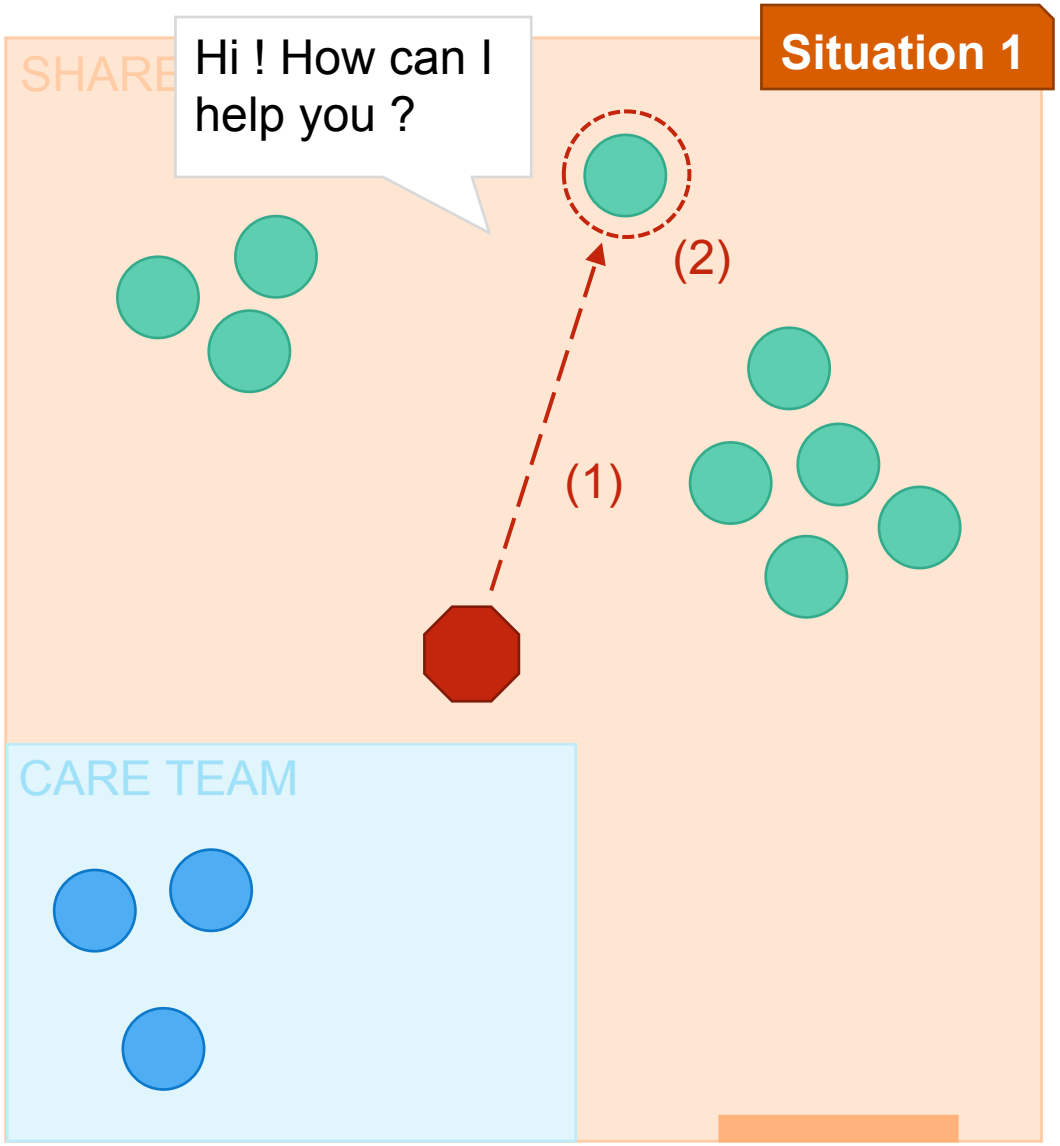
Member of a nursing home care-team

Context



- **Setting**: A shared room with several older adults.
- **Older adults**:
  - Engage in autonomous activities, such as conversing in small groups.
  - Occasionally need assistance, for example:
    - Feeling thirsty
    - Feeling lost or disoriented
    - Seeking information
- **Care Team** (Nurses):
  - Present in the room, gathered in a designated area when not actively engaged.
  - Can bring water, help a patient to move...
- **Robot**:
  - Navigates autonomously within the room.
  - Acts as a member of the care team.
  - Functions as an intermediary between older adults and the care team.
  - Can:
    - Initiate conversations with residents
    - Respond to residents' requests or needs
    - Notify nurses when direct human assistance is required

Scenario



- **Situation 1:** the robot interacts with a resident alone
  - The robot notice someone alone (1), doing nothing special, and go to propose some help (2)
  - [Interaction with the patient]

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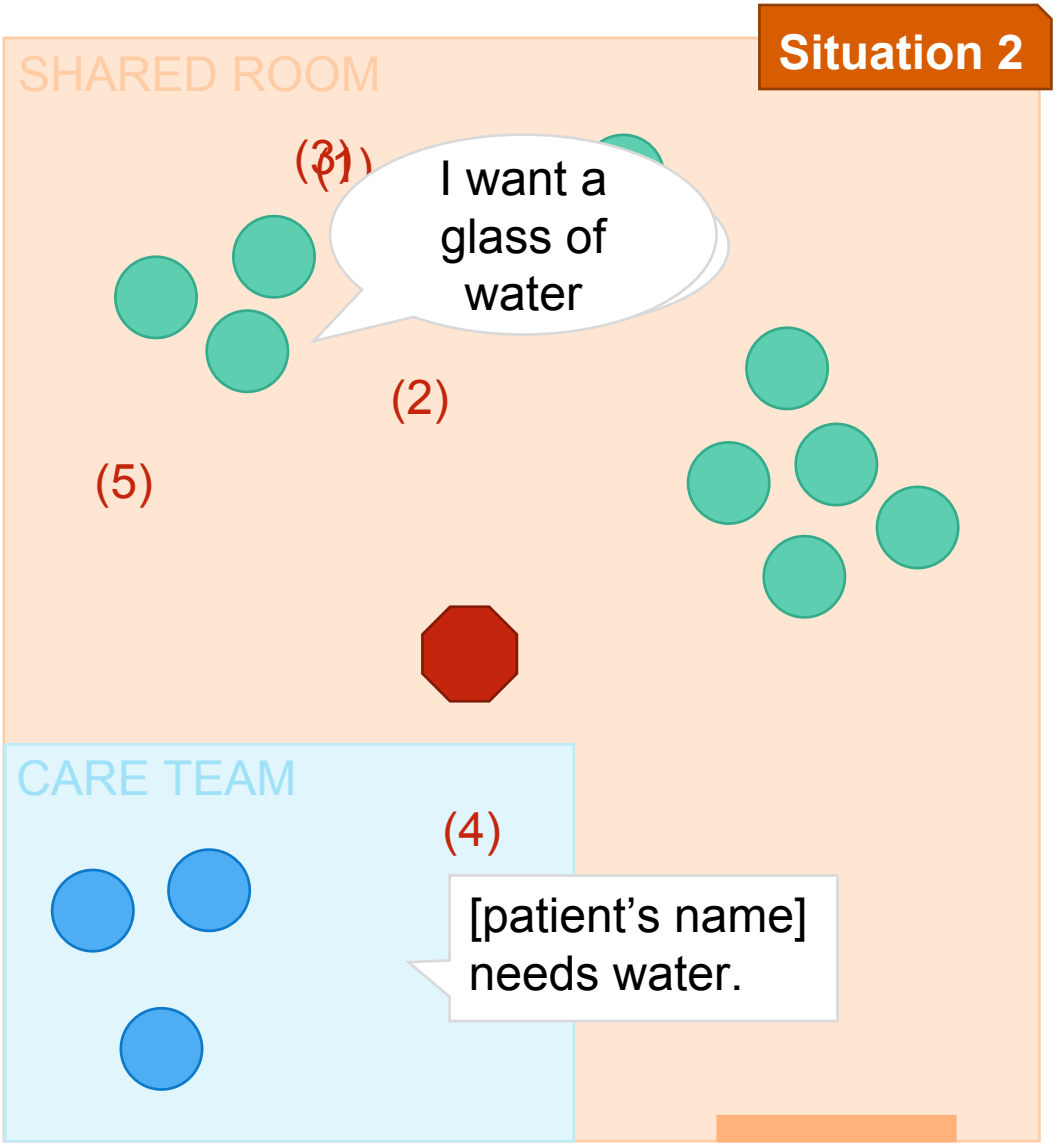
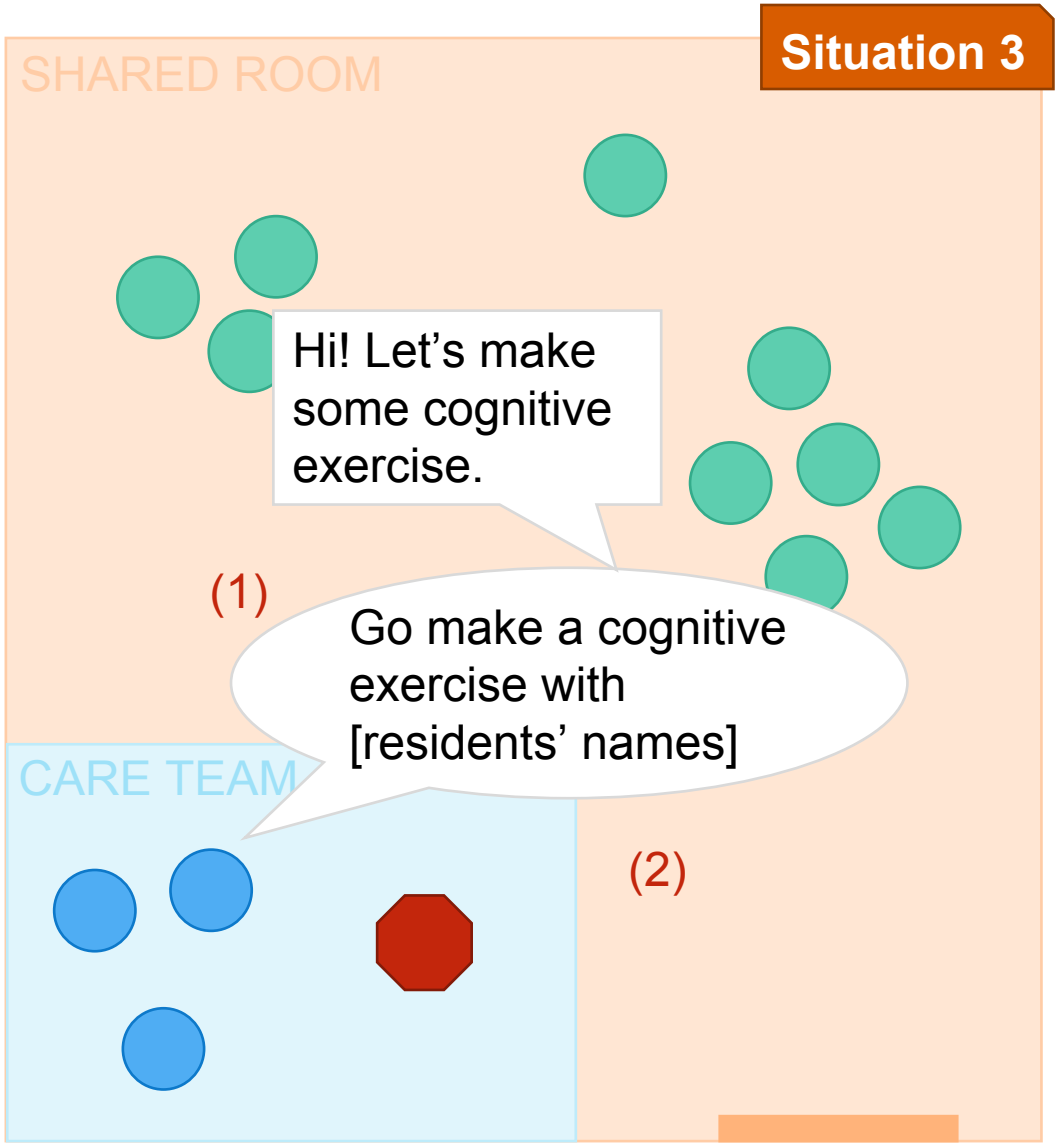


- **Situation 1:** the robot interacts with a resident alone
  - The robot notice someone alone, doing nothing special, and go to propose some help
  - [Interaction with the patient]
- **Situation 2:** a resident calls the robot for help
  - Someone call the robot (1)
  - The robot moves (2)
  - The patient asks for something (e.g. water) (3)
  - The robot returns to the Care Team to notify of the request (4)
  - A nurse take care of the situation (5)

Scenario



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  - A nurse take care of the situation
- **Situation 3:** the care team give the robot a task
  - The Care-Team asks the robot to do cognitive exercise with group of 5 residents (1)
  - The robot go and introduce the game (2)
  - [Cognitive game with residents]

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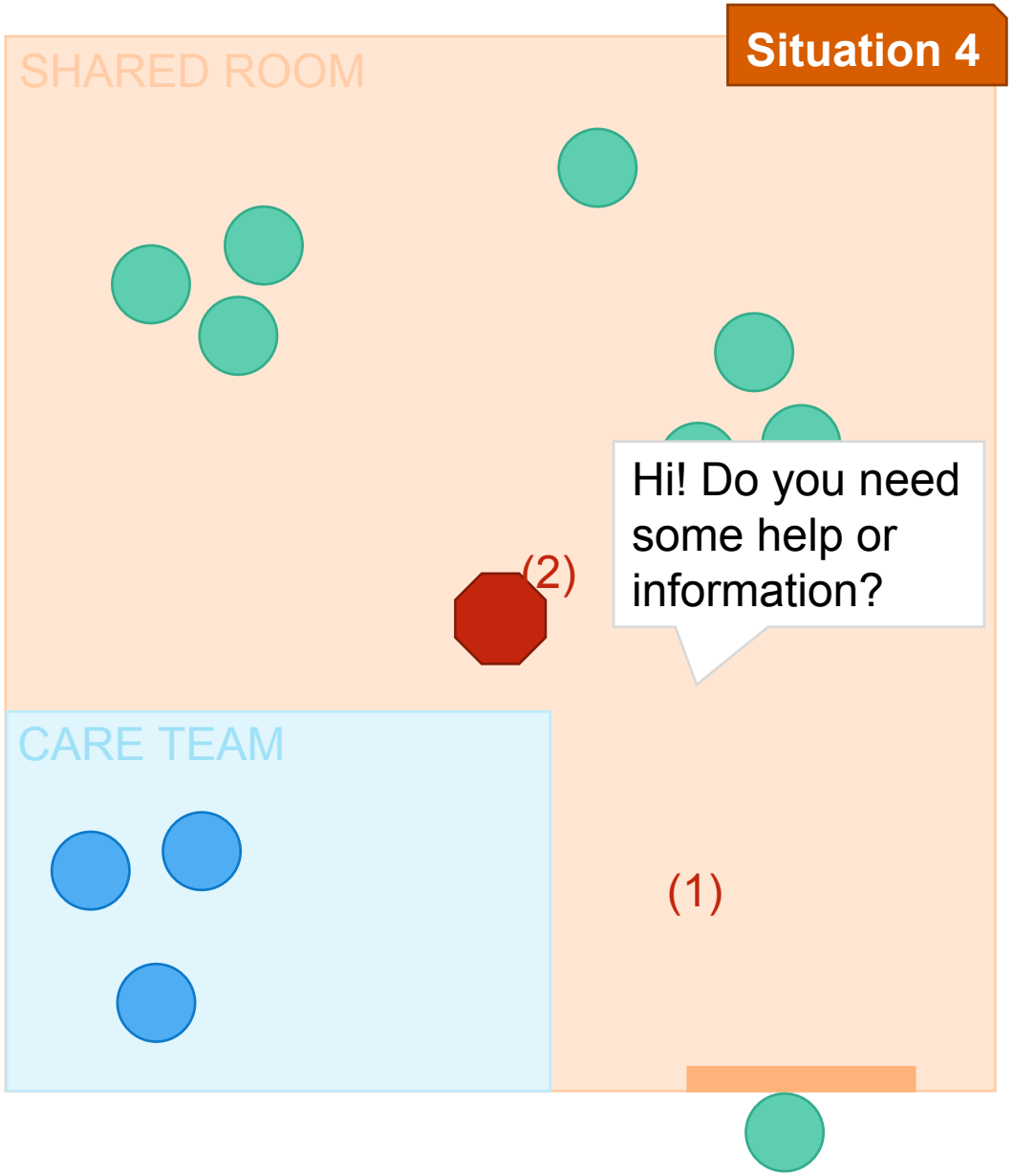

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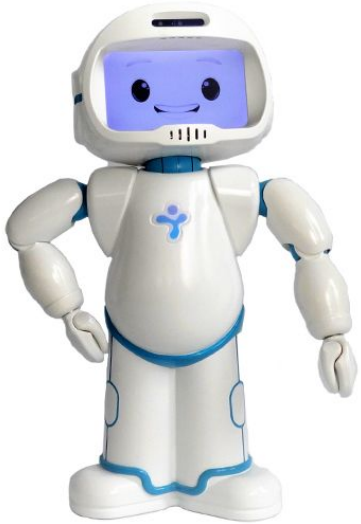
  - [Cognitive game with residents]
- **Situation 4:** the robot notice a resident entering the room
  - A resident enters the room (1)
  - The robot goes to see if he needs help (2)
  - [Interaction with the resident]

Context of our work

Robot used:



QTROBOT: Humanoid robot for social interaction

Abilities:

- Speak and listen
- Move its arms
- Show facial emotions

Small sized (64cm) - friendly face

But: Unable to move

- **User-VLM:** designed for human interactions "user-aware" – adapt the answer to gender, age
- Prompt tuning: adding relevant features using multi-modal data:
  - Audio (frequency, audibility...)
  - Text (Grammar, emotion detection...)
  - Image/Video (facial expression, eyes direction...)

Challenges:

- Ethics and privacy
  - Avoid biases
  - Prevent discrimination
- Keep the user comfortable
  - Tech-ability
  - Low confidence in robotics and IT systems

Situation 1-4: Deal with a resident alone

- **If the resident already does something alone** (reading, playing crosswords...)
  > the robot leaves the resident alone
- **Else,** if the resident could use the robot help (seems lost according to visual cues, just entered the room...)
  > try to initiate a conversation

- **One-to-one** interaction between the robot and a resident
- Interaction **started by the robot**, on its own initiative
  The resident was alone before the robot arrived, and was doing nothing:
  We have to check if:
  - The resident is not lost or confused
  - The resident is comfortable with this interaction

Feature to check in this scenario:

- **Tech-Ability** – if the resident is used or not to interact with the robot
- **Emotion** – sign of distress, anger...
- **Cognitive impairment** – check for confusion, disorientation
- **Physical impairment** – visual clues, determine whether the resident has any mobility impairment

	The resident is lost/confused	**The resident is fine**
The resident is not comfortable with the robot	Go ask the Care Team to help the resident	Leave the resident alone
The resident is comfortable with the robot	Help the resident with his problem, provide user-aware assistance	Propose an activity, or discussion to spend time

Situation 2: Answer to a resident asking for help

- **One-to-one** conversation between the robot and a resident
- Conversation **initiated by the resident**
  The robot needs to:
  - Take note of the resident's problem
  - Determine its ability to solve it itself or not

⚠ If the robot needs help from the care team:

> QTRobot can not move, so it can not **communicate** by directly reporting to them

Same issue if we want the robot to be able to **bring something** to a resident

Tasks that the robot could handle on its own	**Tasks that require an intervention of the Care Team**
▪ Giving information about planning (what time is dinner...) ▪ Provide assistance with conference call, for instance ▪ Propose a cognitive/physical exercise ▪ Play simple games for entertainment ... ?	▪ Helping the resident moving if he is physically impaired (the robot can not take the responsibility if the resident falls). ▪ Bringing something, if it depends on a medical condition the robot is not aware of. ... ?

Situation 3: Complete a task given by the Care Team

- Task given by the team can be from different types, constrained by the robot abilities.
- Notion of success/failure:
  - The robot should report if he was not able to perform the task, and to explain why
  - In case of success, should be able to make a small report, and explain how it went

One-to-one interaction

- Same considerations than before about the resident comfort in the interaction
- The robot has to explain its task to the resident:
  > Adapt to the cognitive abilities, if the task require usage of technology (help for conference call): needs to know how the resident is good with technology

One-to-many interaction

- The robot can not custom its behavior for **every** resident in the interaction
  > adjusts the 'level' of explanations to the individuals who need more time/details
- The robot should now when the residents are addressing to it or to other residents, and avoid cutting them in both cases

Feature to check in this scenario:

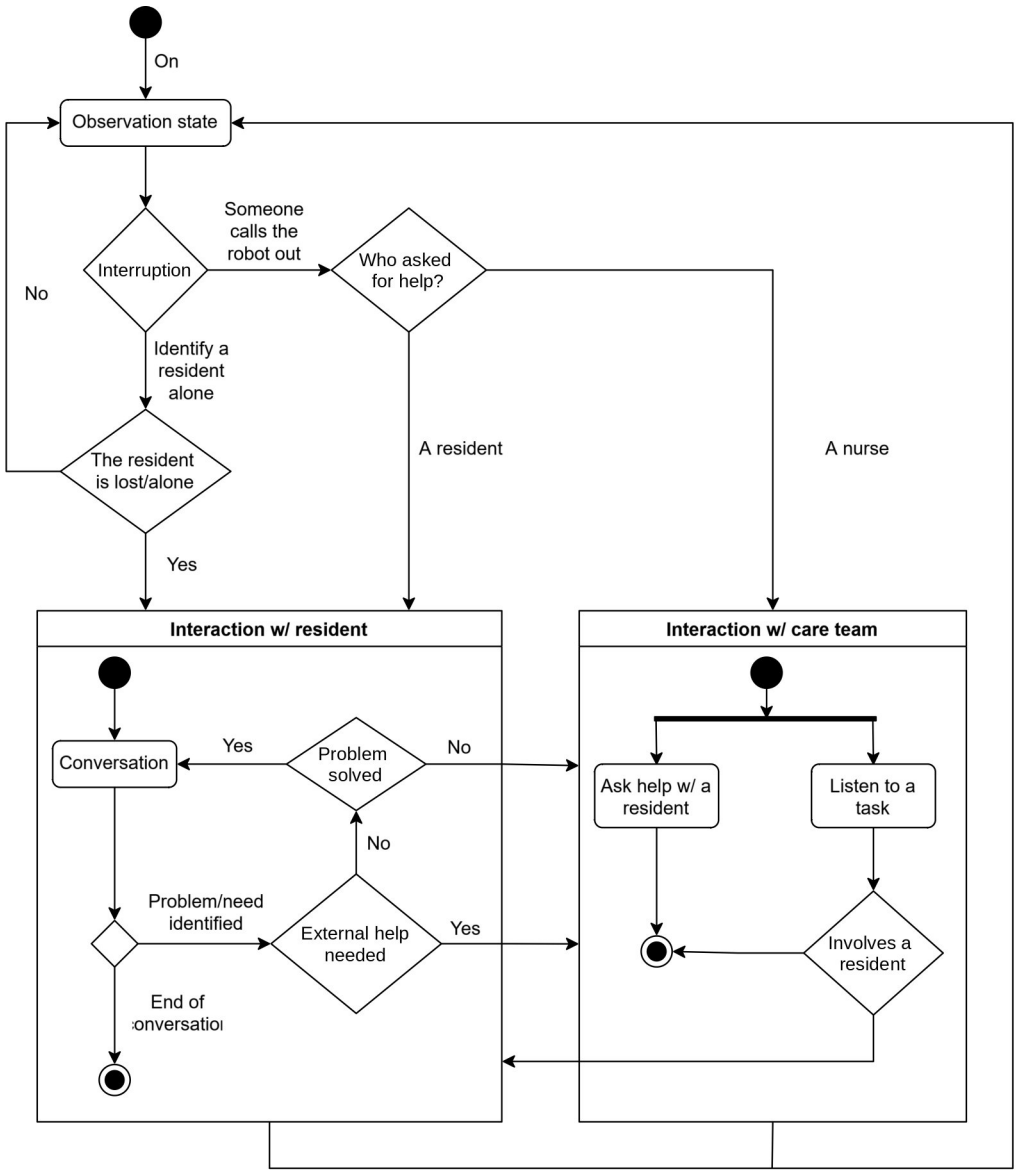
- **Cognitive impairment** – ability to follow an instruction, to remember what was said...
- **Tech-Ability** – level of precision required to explain something related to tech.
- **Emotions** – how much the resident(s) are incline to interact with the robot, to participate in an activity...

Features extraction

- *Technically*, how will the robot extract the features from its interactions

	Features detected by the robot	Audio	Text	Image/Video
Tech-Ability	• Negative emotions like frustration, anxiety • Communication/understanding challenge	Speech rate, volume, pitch, Stutter, filler words...	Pessimism, reluctance Grammatical coherence, asks for repetitions	Facial expression Frowning
Emotions	• Emotional state (happy, sad, angry, neutral...) • Mood instability	Speech tone Crying, trembling Laugh	Positive/negative global sentiment Emotional words	Facial expression Posture, Agitation
Cognitive Impairment	• Confusion or disorientation • Communication/understanding challenge • Slowed response times	Speech rate, Verbal clarity Delayed answers	Incoherence, repetition Difficulty recalling information	?
Physical impairment	• Apparent disabilities • Use of assistive devices • Body language	Apparent pain Breathing difficulties	Explicit mentions of physical impairment	Observation of physical movements, use of assistive devices

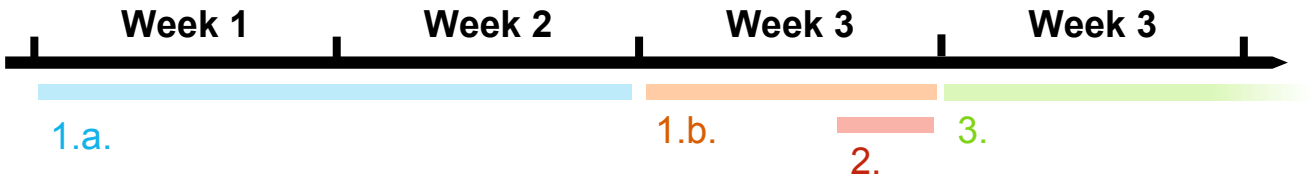
Tasks planning



1. One-to-one interaction with a resident:
   a. One-to-one conversation
   b. Detection of a resident alone

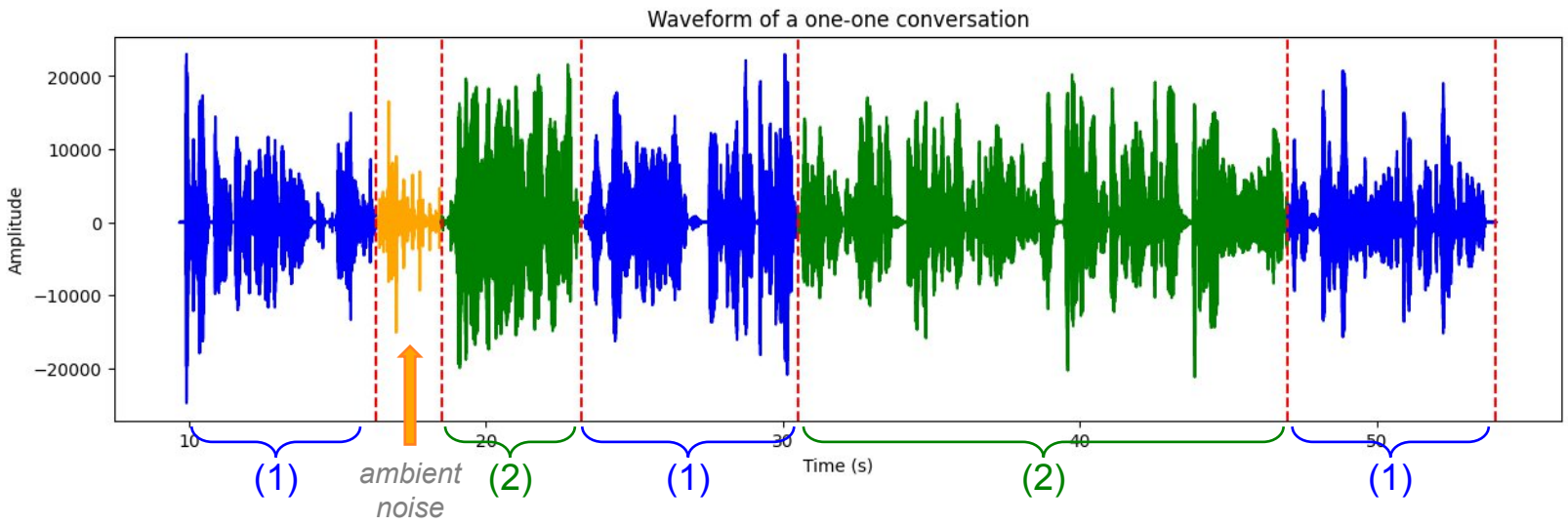
2. Classify individuals:
   nurse, resident, (visitor?)

3. Manage interruptions:
   a. Intervention of a new individual during interaction
   b. Manage one-to-many interactions



1.a. One-to-one interaction: conversation

- Conversation between two people (1) and (2) labeled manually:

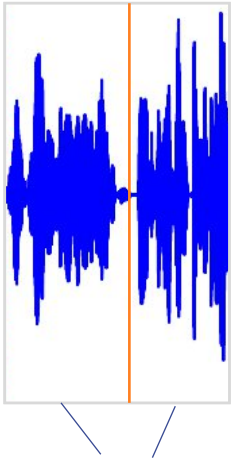


➢ The robot is 'listening' continuously, and needs to:

1. Differentiate voice and other noises (ambient noise...)
2. Perform **speaker diarisation**:
   > Associate every segment with the corresponding individual
   In this case, one of the twp voices will be its own

It can separate the audio in **segments,** marking a separation when it can't hear voice for a certain time (arbitrary threshold)

➢ *Detect stop and alternate person 1 and 2 at each new segment will not work: a long interaction might me divided in many segments.*



Two different segments, but one intervention

1. Identify human voice:

- Using *fft* (fast Fourier transform)

➢ Boolean var. '`talking`':
True if the robot is currently hearing a human voice

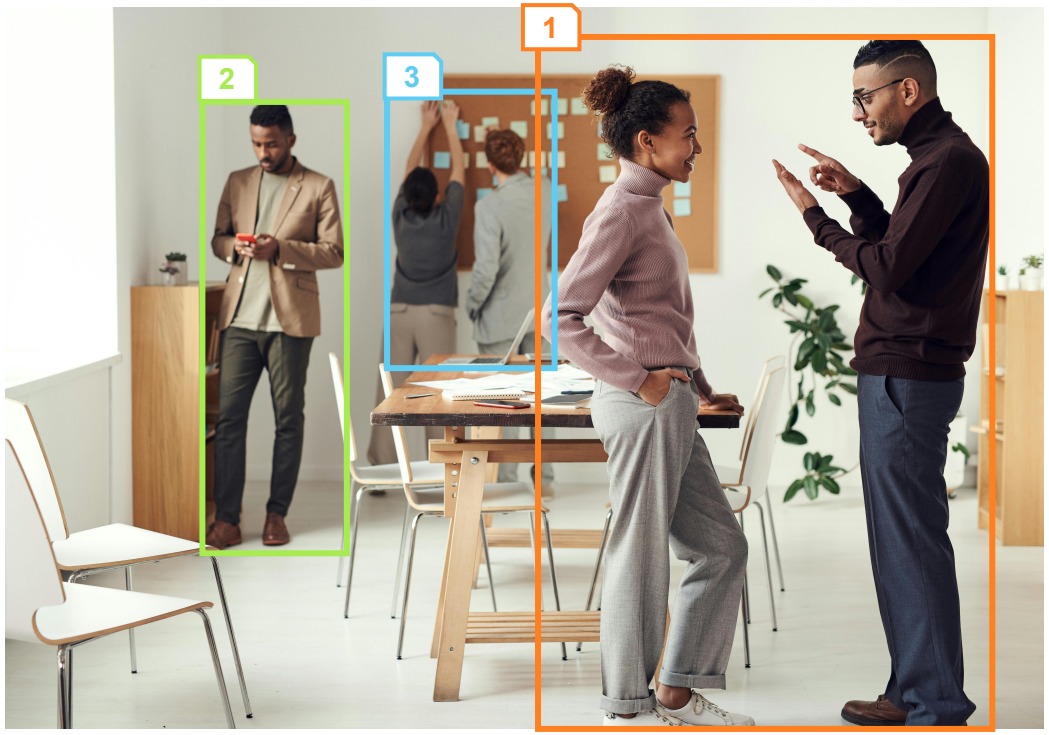
Human voice features:

> dominant frequency between 50 and 400Hz (include male female and children voices)

2. Speaker diarization:

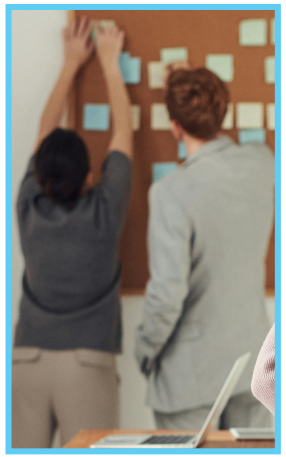
- Popular methods involve Gaussian mixture model and Hidden Markov Model
  > *"A review of speaker diarization: Recent advances with deep learning" - T. Jin Park et.al*

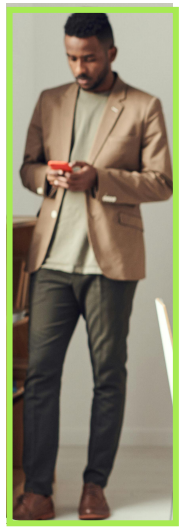
1.b. One-to-one interaction: target detection



1 & 3:
Identify people **in group**

> not concerned by robot interaction





2: Identify individual **alone**
> the robot must determine if he is busy or could you its help

LLM → "a man wearing a suit looking at a smartphone" → Classifier → **> Binary outcome** (whether the robot should start an interaction or no)

Steps:

- Identify individual/groups of individuals
  (*i.e.* determine the bounding boxes on the example)
- Determine if the robot should start an interaction with the individual alone
  > Are they busy or not
- (Differentiate resident and nurses)

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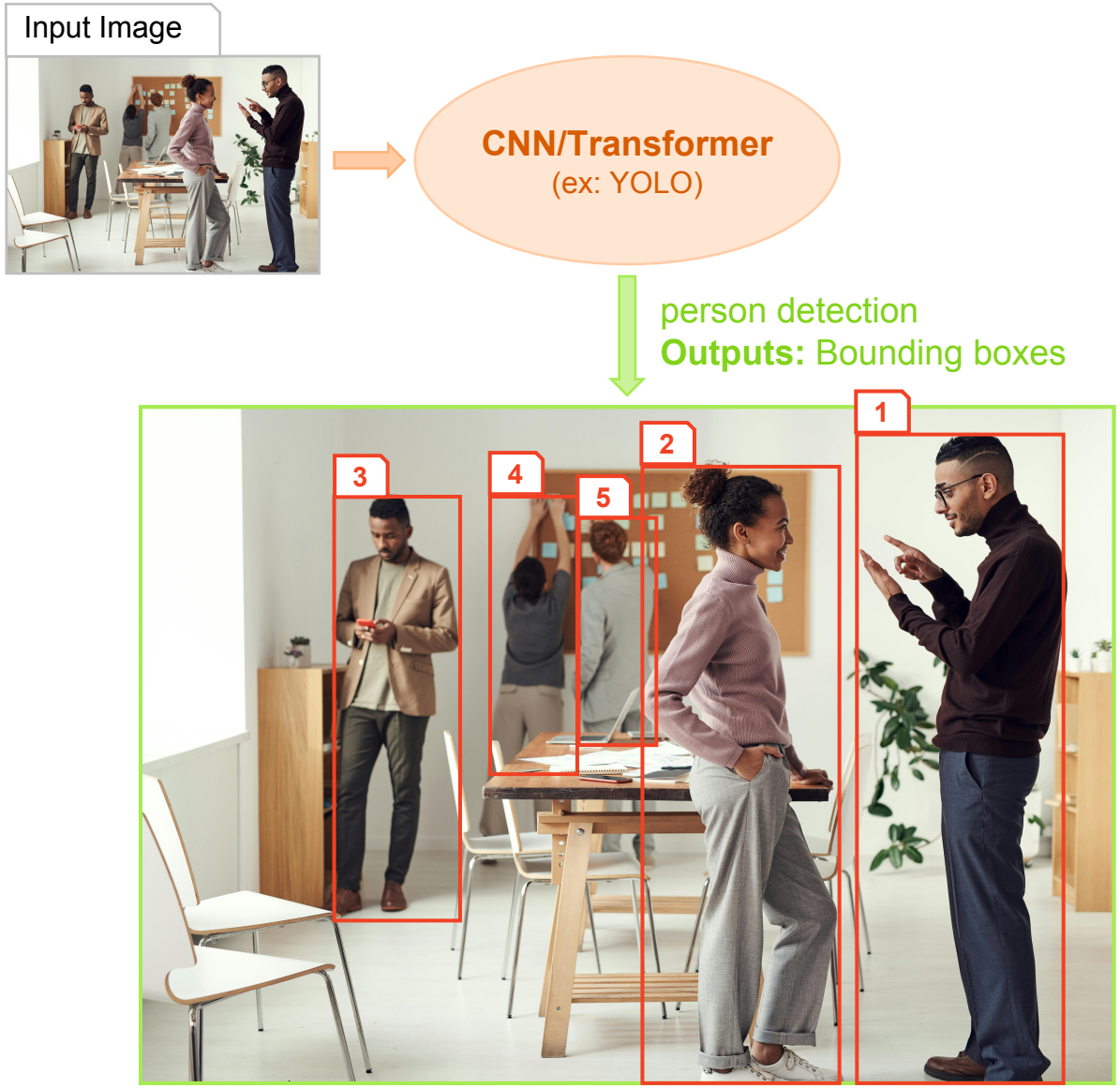
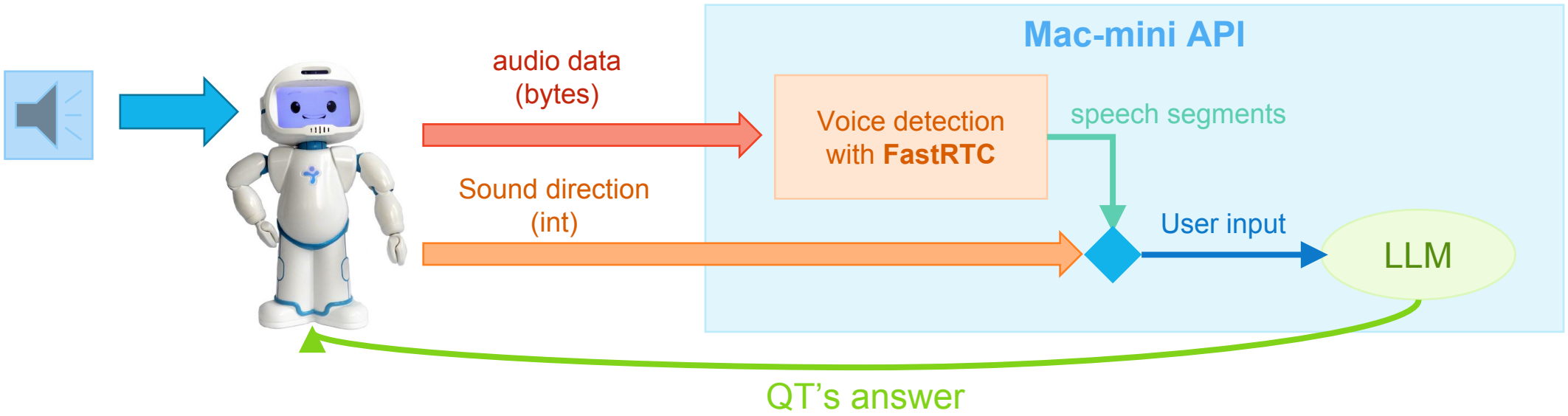


- The distances between the bounding boxes does not allow to predict the groups/alone individuals
  - [2] is closer to [5] than to [1]

- Use depth as other criterion to identify groups
  - Model like MiDaS can provide such heatmaps:

  - Each bounding box could be labeled by the mean of the depth in the area
    (eventually add a step of quick segmentation of the heatmap to avoid background values to be used in the mean)

Conversation with QT (audio)



- QT's microphone does not work well
  - Require speaking close to the 'ears' (Not it front of QT)
  - Works better on certain frequencies when using basic VAD (voice activity detection) like *WebRTC*

➢ Try listening at the audio recorded by QT

Conversation with QT: accessibility and security concerns

- QT should know confidential information about residents
  > Only specific individuals can access this data !

➢ Find a way to prevent QT from revealing sensitive information during conversations

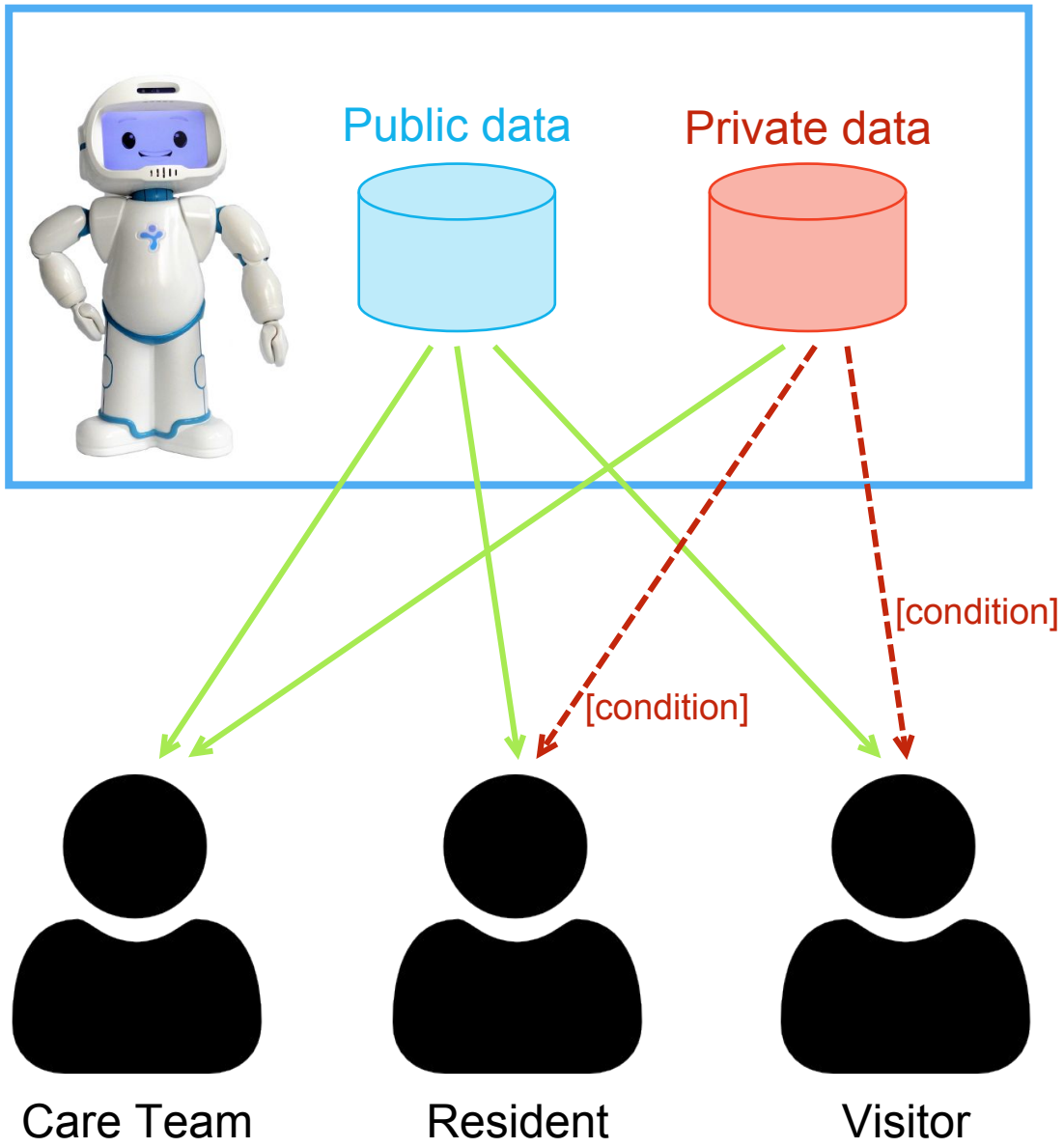
- Save the **sensitive** data in a specific file

- Grant access to people **identified** as members of the Care Team
  > **BUT** risk of mistake in user classification

  OR

  Define an *administrator* mode on QT, unlocked by a **password** known by the Care Team
  > Requires user action

➢ Should the robot be allowed to reveal information to relatives, friends, the concerned resident.... ?



Conversation with QT: Cognitive level adaptation

- Extraction of cognitive features from a conversation through classifiers:
  - ➢ Difficult, ethic concerns...

- Design **"cognitive games"** to infer a cognitive level (decide whether the robot should turn to *"easy mode"* or no)
  - ➢ Compute a score based on the answers of the user (validity, coherence, time needed to answer)...
  - ➢ Rule-based algorithm > Avoid "black box", might be useful for the Care Team if the robot can explain the score attributed

- Cognitive exercises could be basic questions:
  - "What day is today?"
  - "What is your name?"
  - "Do you have a daughter?"

  need identification of the resident